

Exercise 2.25. Prove that $3 \mid (4^n - 1)$ for every $n \in \mathbb{N}$ in two different ways. First, prove it using modular arithmetic. Second, prove it using the fact (which you do not have to prove) that

$$x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + xy^{n-2} + y^{n-1})$$

for any real numbers x and y .

Solution.

We need to prove that

$$3 \mid (4^n - 1)$$

for every $n \in \mathbb{N}$, in two different ways.

First proof: Using modular arithmetic

Since

$$4 \equiv 1 \pmod{3},$$

raising both sides to the n th power gives

$$4^n \equiv 1^n \pmod{3}.$$

Thus,

$$4^n \equiv 1 \pmod{3}.$$

Therefore,

$$4^n - 1 \equiv 0 \pmod{3},$$

which means

$$\boxed{3 \mid (4^n - 1)}.$$

Second proof: Using the given factorization

We are given that

$$x^n - y^n = (x - y) (x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + xy^{n-2} + y^{n-1}).$$

Let

$$x = 4, \quad y = 1.$$

Then

$$4^n - 1^n = (4 - 1) (4^{n-1} + 4^{n-2}(1) + 4^{n-3}(1)^2 + \cdots + 4(1)^{n-2} + 1^{n-1}).$$

Therefore,

$$4^n - 1 = 3 (4^{n-1} + 4^{n-2} + \cdots + 4 + 1).$$

Thus $4^n - 1$ is equal to 3 times an integer. Hence,

$$\boxed{3 \mid (4^n - 1)}.$$

So, by both modular arithmetic and the factorization formula,

$$\boxed{3 \mid (4^n - 1) \text{ for every } n \in \mathbb{N}}.$$